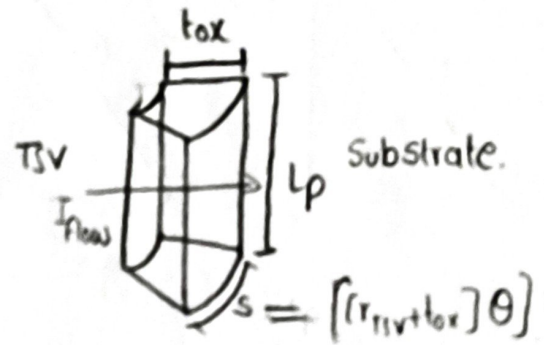
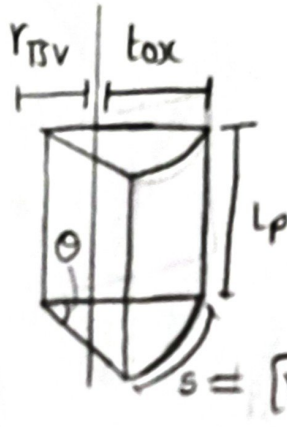
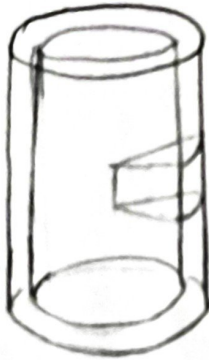


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Questions about topic

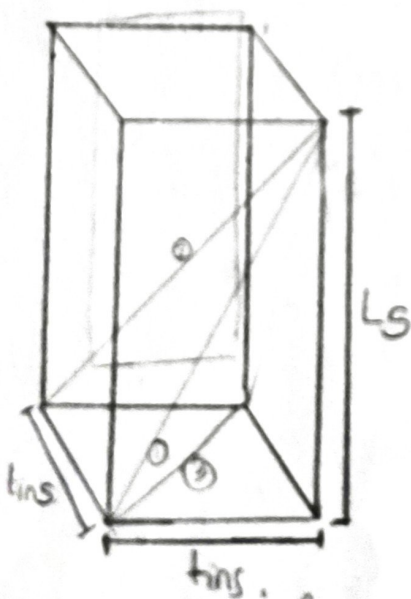
01. Die and wafer (silicon) same?
02. What are the packaging technologies?
03. Why use SiO_2 as separation btw TSV & Si_{sub} ?
04. Higher Integration density?

4 Lshl derivation

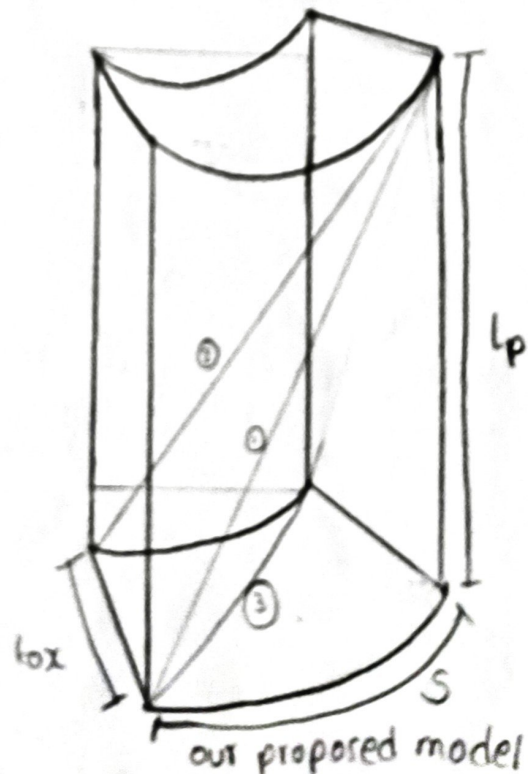


• Assumed Current loop is at infinity.

Similarity btw Rectangular in [Novel paper reference]



A Novel TSV Model by Protenkhalonia model

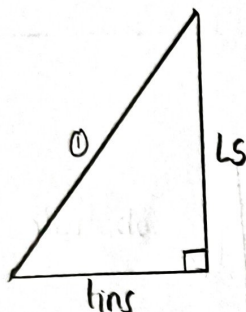


our proposed model

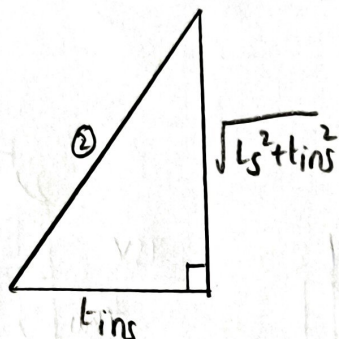
Novel paper

$$L_{sh} = \frac{\mu_0 \mu_{TSV} \cdot t_{ins}}{2\pi} \left[\cosh^{-1} \left[\frac{t_{ins}}{\sqrt{L_s^2 + t_{ins}^2}} \right] - \sqrt{2t_{ins}^2 + L_s^2} + \frac{t_{ins}}{4} + \sqrt{L_s^2 + t_{ins}^2} \right] \quad (H)$$

main field (external)
edge correction
internal inductance
edge correction

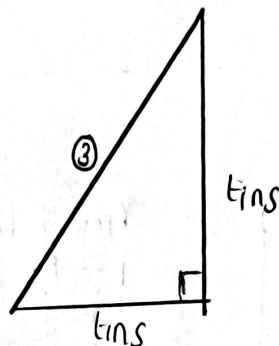


$$① = \sqrt{L_s^2 + t_{ins}^2}$$



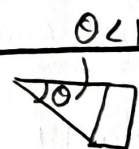
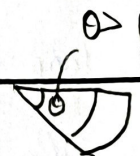
$$\begin{aligned} ② &= \sqrt{t_{ins}^2 + [\sqrt{L_s^2 + t_{ins}^2}]^2} \\ &= \sqrt{t_{ins}^2 + L_s^2 + t_{ins}^2} \end{aligned}$$

$$② = \sqrt{2t_{ins}^2 + L_s^2}$$



$$= \sqrt{t_{ins}^2 + t_{ins}^2}$$

$$③ = \sqrt{2t_{ins}^2}$$



option one. (WRONG)

if $\theta < 1 \text{ rad}$. $\theta \ll \text{small}$ so $L_s \rightarrow S$

if $\theta = \frac{\pi}{4}$ (0.785 rad)

$$\theta = \frac{\pi}{4} \text{ then } J = \frac{\theta L_p}{2\pi r_{TSV}} = \frac{\pi \times (u)}{4 \times 2 \times \pi \times 2.47} = 20.2\%$$

Then

$$L_{sh} = \frac{\mu_0 \mu_{TSV} \cdot t_{ox}}{2\pi} \left[\cosh^{-1} \left[\frac{t_{ox}}{\sqrt{S^2 + t_{ins}^2}} \right] - 2\sqrt{2t_{ox}^2 + S^2} + \frac{t_{ox}}{4} + \sqrt{S^2 + t_{ox}^2} \right]$$

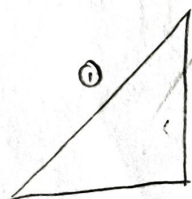
is $S \ll 1$. ≈ 0 .

$$S = L_p (r_{TSV} + t_{ox}) \theta$$

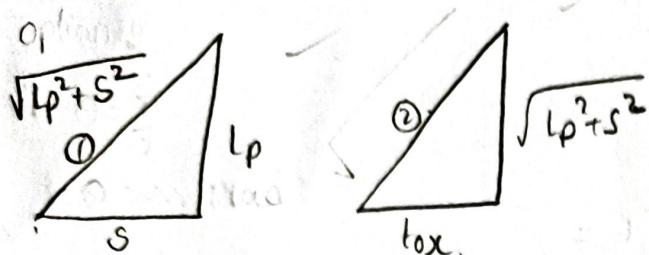
$$= 4 \times 10^{-6} (2.47 + 0.13) \times 10^{-6} \times \frac{3.14}{4}$$

$$= 8.164 \text{ pm}$$

→ very small so almost rectangle.



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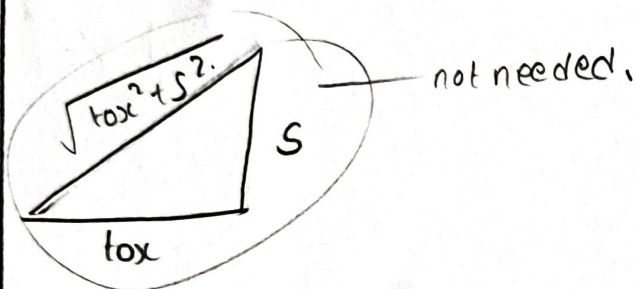


since $s \ll \ll tox$

$\theta = \frac{\pi}{2}$

$$s = 4 \times 10^{-6} [2.47 + 0.13] \times 10^{-6} \frac{2.14}{2} = 16.3 \text{ pm}$$

$$(2) = \sqrt{tox^2 + Lp^2 + s^2}$$



Edge corrections.

$$L_{sh} = \frac{\mu_0 \mu_{TSV} \cdot tox}{2\pi} \left[\cosh^{-1} \left[\frac{s}{\sqrt{tox^2 + Lp^2 + s^2}} \right] - \sqrt{tox^2 + Lp^2 + s^2} + \sqrt{Lp^2 + s^2} \right]$$

Now main field contribution $\cosh^{-1}$ term.

$$\cosh^{-1} \left[\frac{s}{\sqrt{tox^2 + Lp^2 + s^2}} \right] - \sqrt{tox^2 + Lp^2 + s^2} + \left(\frac{s}{4} + \sqrt{Lp^2 + s^2} \right)$$

cross section dimension
diagonal of bar
internal inductance correction term

$$L_{sh} = \frac{\mu_0 \mu_{TSV} tox}{2\pi} \left[\cosh^{-1} \left[\frac{s}{\sqrt{tox^2 + Lp^2 + s^2}} \right] - \sqrt{tox^2 + Lp^2 + s^2} + \frac{s}{4} + \sqrt{Lp^2 + s^2} \right] \quad (41)$$

3, 4, 5, 6, 7, 8, 9, 11